

Chemical Tanker Voyage Economics — Where the Money Actually Is

Round-voyage TCE model, 38,000 dwt IMO II parcel tanker | Sean Burgess | September 2026

Purpose

To establish, for a mid-size deep-sea chemical parcel tanker, which variables actually determine whether a voyage earns its cost of capital — and which of those a commercial team can influence. The model prices a full round voyage (fronthaul laden, backhaul part-laden), builds up every voyage cost line, and converts the result to a time-charter-equivalent day rate.

Base case — US Gulf to ARA round voyage

TCE	Voyage days	Gross freight	Voyage costs	Result vs. benchmark	Break-even rate
\$23,574/day	49.4	\$2.28m	\$1.05m	\$275,406	\$42.77/t

Brent at the trailing-12-month average of \$82.01/bbl, 85% fronthaul and 50% backhaul utilization, normal port time, against a daily vessel-cost benchmark of \$18,000/day (operating expense plus capital recovery).

Findings

Commercial variables outrank fuel. Over plausible ranges, the freight rate swings TCE by \$15,266/day and backhaul utilization by \$13,259/day, against \$5,934/day for a P10-to-P90 move in Brent. Bunkers are the biggest cost line (62% of voyage costs) but rank fifth as a driver of the result.

Cargo intake is the controllable lever. A 10% gain in fronthaul utilization is worth \$3,130/day — about 2.4 times the effect of a 10% move in bunker price, and unlike the oil price it is a decision, not a market outcome. On a parcel tanker, the marginal part cargo is where the margin lives.

Port time is a rate problem. The voyage spends 34% of its days in port. Severe congestion (2x port days) costs \$9,200/day, and lifts the break-even fronthaul rate from \$42.23/t to \$56.35/t at \$80 Brent — a 33% uplift the market does not automatically pay. Berth productivity and laytime terms are commercial terms, not operational details.

Route economics turn on one insurance line. On the ARA-India trade, Suez earns \$21,396/day against \$16,496/day around the Cape. Remove the illustrative war-risk premium (0.25% of hull value per transit) and Suez gains \$3,009/day. Distance does not decide the routing; the premium does.

Sensitivity ranking — TCE swing over the plausible range of each input

Input	Range tested	TCE at low	TCE at high	Swing \$/day	Range set by
Freight rate (market)	0.91x to 1.25x	\$19,533	\$34,799	\$15,266	<i>BLS PPI YoY P10/P90</i>
Backhaul utilization	25% to 75%	\$16,945	\$30,204	\$13,259	<i>judgment band</i>
Port time (congestion)	0.75x to 1.75x	\$26,505	\$17,305	\$9,200	<i>scenario band</i>
Fronthaul utilization	70% to 95%	\$18,105	\$27,259	\$9,154	<i>judgment band</i>
Bunker price (Brent)	\$68 to \$105	\$25,833	\$19,899	\$5,934	<i>Brent 5-yr P10/P90</i>
Port disbursements	0.75x to 1.25x	\$24,851	\$22,298	\$2,553	<i>judgment band</i>
Service speed	12.0 kn to 14.5 kn	\$24,473	\$22,463	\$2,010	<i>operating range</i>
EU carbon price (EUA)	EUR 50 to EUR 110	\$24,410	\$22,404	\$2,007	<i>judgment band</i>

All other inputs held at base case in each row. Every case is a complete re-run of the voyage, not a linear approximation.

Recommendation

Prioritize commercial effort in the order the sensitivity implies: rate negotiation and cargo intake first, port and laytime terms second, bunker and speed optimization third. Bunker hedging protects a real but secondary exposure; it does not substitute for filling the ship. Re-optimize speed per fixture rather than by policy — the TCE-maximizing speed in this model moves by 2.5 knots across the plausible bunker range. For any trade touching the EU, carry the ETS cost (\$123,936 on this voyage, about \$2,508/day) in the rate, not in the margin.

Assumptions and limitations — read before using any number above

Simulated and estimated inputs. Freight rates, port costs, port days, vessel particulars and the crude-to-bunker conversion are analyst assumptions, not observed market data. Canal tolls are built up from published Panama Canal Authority and Suez Canal Authority schedules, and US Gulf and ARA port dues from Port Houston Tariff No. 14 and the Port of Rotterdam seaport-dues schedule; Brent and the freight index are public series. Port ancillaries (pilotage, tugs, agency) and the Far East and India calls remain estimates. No public chemical-tanker rate assessment exists (Baltic and Clarksons are subscription-only), so base rates were calibrated to a plausible TCE and the BLS deep-sea freight PPI was used only to size the plausible range. Vessel-level congestion data (USACE LPMS) is not publicly available, so port time is modelled as scenario bands. Every assumption is a labelled, editable input cell in the accompanying workbook; the conclusions concern the structure of voyage economics and are robust to the level of these inputs, while the dollar figures are illustrative.

Independent analysis using publicly available data. Not affiliated with or endorsed by any organization named. Model: chemical_tanker_voyage_model.xlsx (five scenario switches, full cost build-up, tornado, break-even curves); code: analysis.py; sources: data/README.md.